🧪 EduTrack — Test Cases Document

Academic Management Platform

VTU Internship 2026 — Team 15

Testing Overview

TEST TYPE	TOOL	TOTAL TESTS
Unit Tests (Service Layer)	JUnit 5 + Mockito	22 tests
Controller Tests (API Layer)	Spring MockMvc	8 tests
End-to-End Tests (Browser)	Playwright	43 tests
API Tests	Postman	26 requests
Total		**99 tests**

Part 1 — JUnit Unit Tests (AuthServiceTest)

Module: Student Login

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-SL-01	Valid student login	email: student@test.com, password: Test@123	success: true, role: student	Pass
TC-SL-02	Wrong password	email: student@test.com, password: wrongpass	success: false, message: "Invalid email or password"	Pass
TC-SL-03	Email not found	email: nobody@test.com, password: Test@123	success: false	Pass

Module: Teacher Login

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-TL-01	Valid teacher login	email: teacher@test.com, password: Test@123	success: true, role: teacher	Pass
TC-TL-02	Wrong password	email: teacher@test.com, password: wrongpass	success: false	Pass

Module: Admin Login

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-AL-01	Valid admin login	email: admin@edutrack.com, password: Admin@123	success: true, role: admin	Pass
TC-AL-02	Wrong password	email: admin@edutrack.com, password: wrongpass	success: false, message: "Invalid email or password"	Pass
TC-AL-03	Email not found	email: ghost@test.com, password: Admin@123	success: false	Pass

Module: Change Password

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-CP-01	Admin correct current password	adminId: 1, current: Admin@123, new: NewPass@123	success: true, message: "Password changed successfully"	Pass
TC-CP-02	Admin wrong current password	adminId: 1, current: wrongpass, new: NewPass@123	success: false, message: "Current password is incorrect"	Pass
TC-CP-03	Admin not found	adminId: 99, current: Admin@123, new: NewPass@123	success: false, message: "Admin not found"	Pass
TC-CP-04	Student correct current password	studentId: 1, current: Test@123, new: NewPass@123	success: true	Pass
TC-CP-05	Student wrong current password	studentId: 1, current: wrongpass, new: NewPass@123	success: false, message: "Current password is incorrect"	Pass

Module: Student Registration

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-SR-01	New email registration	email: new@test.com	success: true	Pass
TC-SR-02	Duplicate email	email: student@test.com (already exists)	success: false, message: "Email already registered"	Pass

Part 2 — JUnit Unit Tests (OtpServiceTest)

Module: OTP Validation

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-OTP-01	Valid OTP within expiry	email: test@test.com, otp: 123456, expiry: +5 min	true	Pass
TC-OTP-02	Wrong OTP	email: test@test.com, otp: 999999	false	Pass
TC-OTP-03	Expired OTP	email: test@test.com, otp: 123456, expiry: -15 min	false	Pass
TC-OTP-04	Already used OTP	email: test@test.com, otp: 123456, used: true	false	Pass
TC-OTP-05	No OTP record found	email: ghost@test.com	false	Pass
TC-OTP-06	OTP marked as used after success	email: test@test.com, otp: 123456	used = true, save() called once	Pass

Module: OTP Generation

TC ID	TEST CASE	INPUT	EXPECTED OUTPUT	STATUS
TC-OTP-07	OTP saved to repository	email: test@test.com	otpRepo.save() called once	Pass
TC-OTP-08	Old OTP deleted before new one	email: test@test.com	otpRepo.deleteByEmail() called once	Pass

Part 3 — Controller Tests (AuthControllerTest)

Module: Admin Login API

TC ID	TEST CASE	ENDPOINT	INPUT	EXPECTED HTTP	EXPECTED RESPONSE	ST
TC-API-01	Valid admin login	POST /api/auth/login/admin	email: admin@edutrack.com, password: Admin@123	200 OK	success: true, role: admin	P
TC-API-02	Wrong admin password	POST /api/auth/login/admin	email: admin@edutrack.com, password: wrongpass	200 OK	success: false, message: "Invalid email or password"	P

Module: Student Login API

TC ID	TEST CASE	ENDPOINT	INPUT	EXPECTED HTTP	EXPECTED RESPONSE	STA
TC-API-03	Valid student login	POST /api/auth/login/student	email: student@test.com, password: Test@123	200 OK	success: true, role: student	Pa
TC-API-04	Wrong student password	POST /api/auth/login/student	email: student@test.com, password: wrongpass	200 OK	success: false	Pa

Module: Teacher Login API

TC ID	TEST CASE	ENDPOINT	INPUT	EXPECTED HTTP	EXPECTED RESPONSE	STATU
TC-API-05	Valid teacher login	POST /api/auth/login/teacher	email: teacher@test.com, password: Test@123	200 OK	success: true, role: teacher	Pass

Module: Change Password API

TC ID	TEST CASE	ENDPOINT	INPUT	EXPECTED HTTP	EXPECTED RESPONSE	STATUS
TC-API-06	Correct current password	POST /api/auth/change-password/admin	id: 1, current: Admin@123, new: NewPass@123	200 OK	success: true	Pass
TC-API-07	Missing required fields	POST /api/auth/change-password/admin	id: 1 only	200 OK	success: false	Pass

Part 4 — Playwright E2E Tests

Module: Authentication

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-01	Landing page loads	Open app URL	EduTrack text visible	Pass
TC-E2E-02	Login page shows 3 role buttons	Navigate to login	Student, Teacher, Admin buttons visible	Pass
TC-E2E-03	Switching roles clears email	Type email, switch role	Email field cleared	Pass
TC-E2E-04	Admin login success	Enter valid admin credentials	Admin Portal visible	Pass
TC-E2E-05	Wrong password shows error	Enter wrong password	"Invalid email or password" shown	Pass
TC-E2E-06	Empty fields shows error	Click login without filling	"Please fill in all fields" shown	Pass
TC-E2E-07	Admin logout	Click Logout	Redirected to landing page	Pass

Module: Admin Student Management

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-08	Students page loads	Login as admin, click Students	Student Management visible	Pass
TC-E2E-09	Search filters students	Type "naveen" in search	Matching students shown	Pass
TC-E2E-10	Edit student modal opens	Click Edit on a student	Edit Student modal visible	Pass
TC-E2E-11	Email is read-only	Open edit modal	No email input field	Pass
TC-E2E-12	No department/year/semester dropdowns	Open edit modal	No select dropdowns	Pass
TC-E2E-13	Cancel closes modal	Click Cancel	Modal disappears	Pass

Module: Admin Teacher Management

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-14	Teachers page loads	Login as admin, click Teachers	Teacher Management visible	Pass
TC-E2E-15	Edit teacher modal opens	Click Edit on a teacher	Edit Teacher modal visible	Pass
TC-E2E-16	Department is read-only	Open edit modal	No department dropdown	Pass
TC-E2E-17	Email is read-only	Open edit modal	No email input	Pass
TC-E2E-18	Cancel closes modal	Click Cancel	Modal disappears	Pass

Module: Admin Settings

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-19	Settings page loads	Login as admin, click Settings	Personal Information visible	Pass
TC-E2E-20	System tab visible for admin	Open settings	System tab visible	Pass
TC-E2E-21	Change password modal opens	Click Security, then Change Password	Modal visible	Pass
TC-E2E-22	Wrong current password shows error	Enter wrong current password	Error message shown	Pass
TC-E2E-23	Cancel closes modal	Click Cancel	Password inputs gone	Pass

Module: Forgot Password

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-24	Forgot password page loads	Click Forgot password?	3 role options visible	Pass
TC-E2E-25	Shows 3 role options	Open forgot password	3 role buttons visible	Pass
TC-E2E-26	Switching role clears email	Type email, switch role	Email cleared	Pass
TC-E2E-27	Non-existent email shows error	Enter ghost@nowhere.com	"No account found" shown	Pass
TC-E2E-28	Wrong role for email shows error	Admin email as Teacher role	"No account found" shown	Pass
TC-E2E-29	Empty email shows error	Click Send OTP without email	"Please enter your email" shown	Pass
TC-E2E-30	Back to login works	Click Sign in	Login page shown	Pass

Module: Student Portal

TC ID	TEST CASE	STEPS	EXPECTED RESULT	STATUS
TC-E2E-31	Student dashboard loads	Login as student	Student Portal visible	Pass
TC-E2E-32	Nav items visible	Login as student	My Courses, Attendance, Assignments, My Marks visible	Pass
TC-E2E-33	Settings page loads	Click Settings	Personal Information visible	Pass
TC-E2E-34	No System tab for student	Open settings	System tab not visible	Pass

Module: Backend API (Direct)

TC ID	TEST CASE	ENDPOINT	EXPECTED RESPONSE	STATUS
TC-API-08	Admin login returns success	POST /api/auth/login/admin	success: true, role: admin	Pass
TC-API-09	Wrong password returns failure	POST /api/auth/login/admin	success: false	Pass
TC-API-10	OTP rejects unknown email	POST /api/otp/forgot-password	success: false, "No account found"	Pass
TC-API-11	OTP rejects wrong role	POST /api/otp/forgot-password	success: false	Pass
TC-API-12	GET students returns array	GET /api/students	Array of students	Pass
TC-API-13	GET teachers returns array	GET /api/teachers	Array of teachers	Pass
TC-API-14	Wrong current password fails	POST /api/auth/change-password/admin	success: false, "incorrect"	Pass
TC-API-15	Invalid OTP rejected	POST /api/otp/verify	success: false	Pass
TC-API-16	System settings returns data	GET /api/system/settings	activeYear, maintenanceMode present	Pass

Test Summary

MODULE	JUNIT	PLAYWRIGHT	TOTAL
Student Login	3	0	3
Teacher Login	2	0	2
Admin Login	3	2	5
Change Password	5	2	7
Registration	2	0	2
OTP Validation	8	4	12
Admin Student Mgmt	0	6	6
Admin Teacher Mgmt	0	5	5
Admin Settings	0	5	5
Student Portal	0	4	4
Backend API	8	9	17
Total	**31**	**43**	**74**